1.06 Quadratics: Introduction

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1: Solving Quadratic Equations

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$$x^2 + 5x = 6$$

There are three ways of solving a quadratic equation.
One is by *completing the square*, which we'll do later.
What are the other two ways?

By factorisation

Using the quadratic formula.

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1: Solving Quadratic Equations

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Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

$$(x-1)^2 = 5$$

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Test Your Understanding

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1 Solve $(x+3)^2 = x+5$ using factorisation.

2 Solve $(2x+1)^2 = 5$

3 Solve $\sqrt{x+3} = x-3$

4 Solve $2x + \sqrt{x} - 1 = 0$

"Use of Technology":
The 'equation' mode on your calculator will solve quadratics in the form $ax^2+bx+c=0$. When you're asked for the 'order', use 2 (we'll see why later).

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Exercise 2A/2B Pure Y1 Pages 20-22

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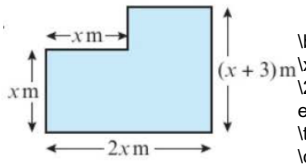
3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

\
x
\
x

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.



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5 Solve the equation $5x + 3 = \sqrt{3x+7}$.

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2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

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Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1 (i) Use the substitution $\sqrt{x} = y$ $(y \geq 0)$ to find the real root of the equation $x + 3\sqrt{x} - \frac{1}{2} = 0$.

(ii) Find all real roots of the following equations:

(a) $x + 10\sqrt{x + 2} - 22 = 0$;

(b) $x^2 - 4x + \sqrt{2x^2 - 8x - 3} - 9 = 0$.

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2: Completing the Square

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"Completing the square" means putting a quadratic in the form $(x+a)^2+b$ or $a(x+b)^2+c$

The **underlying reason we do this is because x only appears once in the expression** (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

If we have a completed square:

$$(x+4)^2-7=0$$

we saw at the start of the chapter how we could rearrange to make x the subject.
Indeed using the quadratic formula is actually solving the quadratic by completing the square – it's just someone has done the work for us already!

b. Sketching Quadratics

We'll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to 'integrate' expressions like:

$$\int \frac{1}{x^2-4x+5}\,dx$$

(you will cover integration later this module)

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Completing the Square Recap

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Expand:

$(x+\mathbf{9})^2 = x^2+$

$(x-\mathbf{5})^2 = x^2-$

What do you notice about the relationship between the bold numbers?

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Completing the Square

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Express $2x^2 + 12x + 7$ in the form $a(x+b)^2 + c$

$= 2\left(x\right.$

$= 2\left(\left(\right.\right.$

$= 2(x$

$= 2(x$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x+c)^2$

$= -3x^2$

$= -3\left(x\right.$

$= -3\left(\left(\right.\right.$

$= -3(x$

$= 8 - 3($

It may help to write in the form $ax^2 + bx + c$ first.

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Test Your Understanding

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Express $3x^2 - 18x + 4$ in the form $a(x+b)^2 + c$

$= 3\left(x\right.$

$= 3\left(\right.$

$= 3(x$

$= 3(x$

Express $20x - 5x^2 + 3$ in the form $a - b(x+c)^2$

$= -5$

$= -5$

$= -5$

$= -5$

$= 23$

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Solving by Completing the Square

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Solve the equation:

$$3x^2 - 18x + 4 = 0$$

x^2

$(x$

$(x$

x

x

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Exercise 2C/2D Pure Y1 Page 23-24

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1 Complete the square for the expressions: **a** $x^2 + 4x$ **b** $x^2 - 6x$ **c** $x^2 - 16x$ **d** $x^2 + x$ **e** $x^2 - 14$

2 Complete the square for the expressions: **a** $2x^2 + 16x$ **b** $3x^2 - 24x$ **c** $5x^2 + 20x$ **d** $2x^2 - 5x$ **e** $8x - 2x^2$

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4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\mathrm{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \mathrm{and} B$ are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

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Proving the Quadratic Formula

18

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$ax^2 + bx + c = 0$

$x^2 + \frac{b}{a}$

$\left(x + \frac{b}{2}\right.$

$\left(x + \frac{b}{2}\right.$

$\left(x + \frac{b}{2}\right.$

$x + \frac{b}{2a}$

$x = \frac{-b}{}$

Just use exactly the same method as you usually would!